



Customer Shopping Behavior Analysis

Uncovering insights from 3,900 customer purchases to drive data-driven business decisions



Project Overview



Dataset

3,900 customer purchases across multiple product categories



Objective

Identify spending patterns, trends, preferences, and shopping habits



Impact

Improve sales, marketing strategies, and customer engagement

Dataset Summary

3,900 Records

18 columns of customer data

- Customer demographics
- Purchase information
- Shopping behavior



Demographics

Age, Gender, Location,
Subscription Status



Purchase Info

Item, Category, Amount, Season,
Size, Color

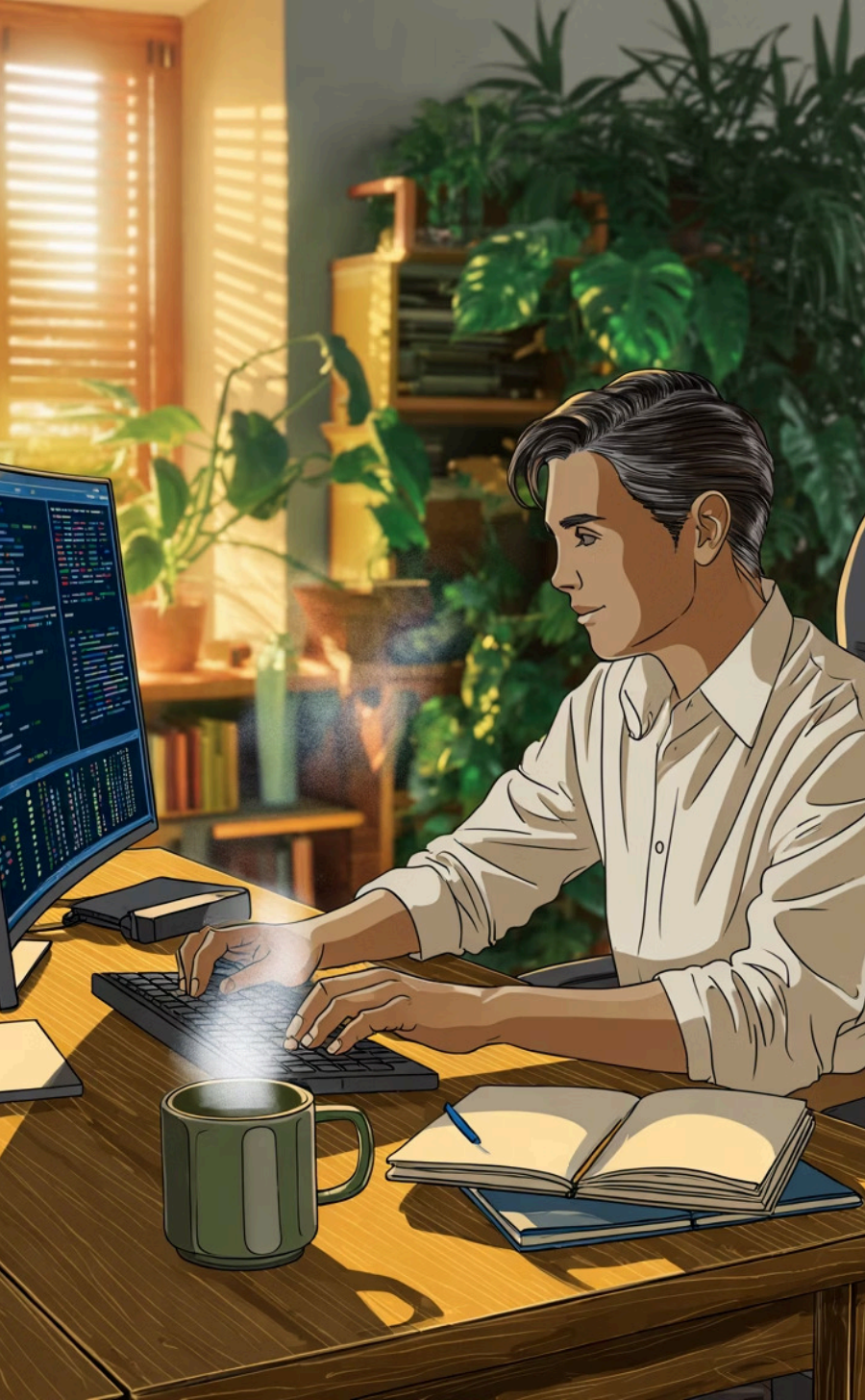


Behavior

Discount, Promo, Frequency,
Rating, Shipping

Data Quality

37 missing values handled in Review Rating column



Data Preparation & Cleaning

01

Data Loading

Imported dataset using pandas

02

Initial Exploration

Checked structure and summary statistics

03

Missing Values

Filled Review Rating using median by category

04

Column Formatting

Standardized to snake_case for consistency

Feature Engineering

Age Groups

Categorized customers into age ranges

Purchase Frequency

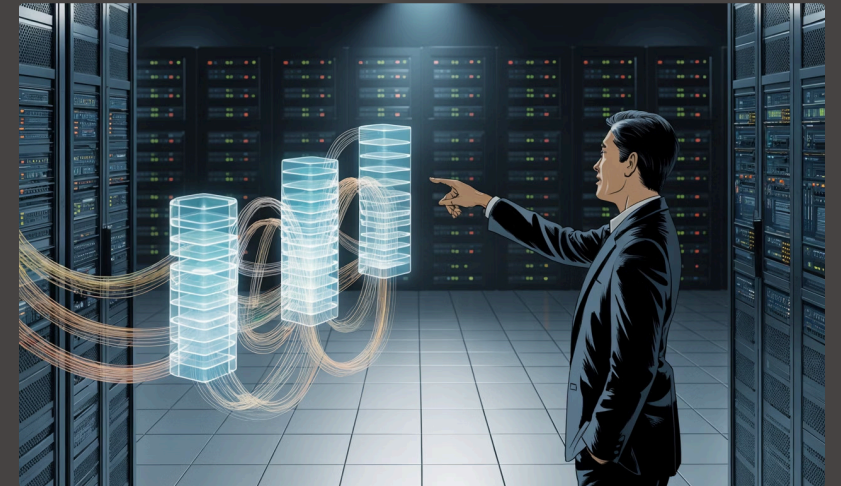
Created
`purchase_frequency_days`
feature

Data Validation

Removed redundant
`promo_code_used` column

PostgreSQL Integration

Imported cleaned dataset for
SQL analysis



SQL Analysis Highlights



Revenue by Gender

Compared total revenue generated by male and female customers



High-Spending Discount Users

Identified customers who applied discounts yet purchased above average



Top Products by Rating

Top 5 products with highest average customer review ratings

Customer Segmentation

Subscribers vs. Non-Subscribers

Compared average purchase amount and total revenue by subscription status

Discount-Dependent Products

Identified 5 products with highest percentage of discounted purchases

Customer Segments

Classified as New, Returning, or Loyal based on purchase history



Additional Insights

- Shipping Type Comparison

Analyzed spending by Standard vs. Express shipping preferences

- Top Products per Category

Listed most purchased products within each category

- Repeat Buyers & Subscriptions

Customers with >5 purchases more likely to subscribe

- Revenue by Age Group

Calculated total revenue contribution of each age group





Power BI Dashboard



Visualizations

Revenue and sales by category



DAX Measures

KPI and sales analysis



Slicers

Dynamic filtering



Key Takeaways



Data-Driven Decisions

Comprehensive analysis supports strategic business choices



Revenue Insights

Gender, age, and subscription impact on spending patterns



Customer Segments

Clear classification enables targeted marketing strategies